A Fast Method for Inferring High-Quality Simply-Connected Superpixels

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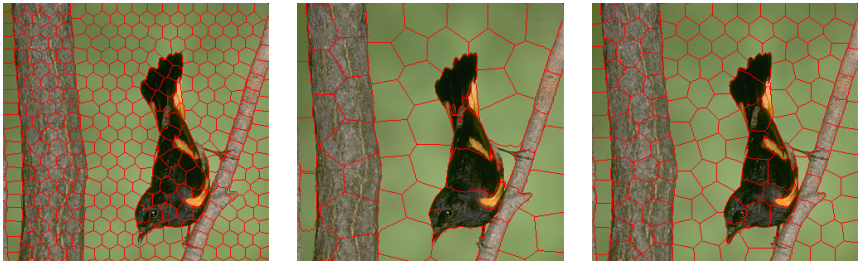
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   - Modifying a method by Chang, Wei and Fisher
   - Improved modeling
   - Faster Inference
3. Results and comparisons



Superpixels[1]

Coherent image patches; boundaries well aligned with image edges (but not too irregular)



Results of the proposed method for different values of K

K: #superpixels

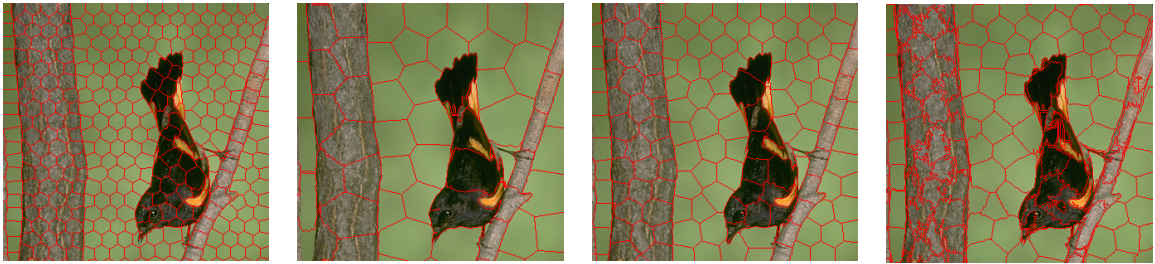
N: #pixels

$K \ll N$

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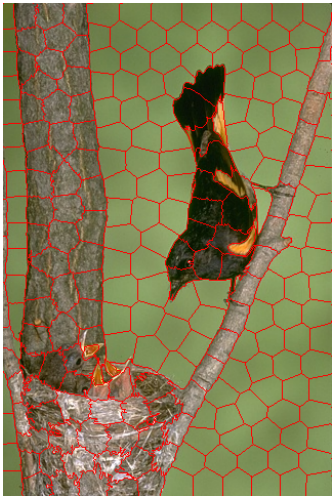
Superpixels: a Compact Intermediate Representation

A useful pre-processing for many tasks

Image



Borders



Average colors



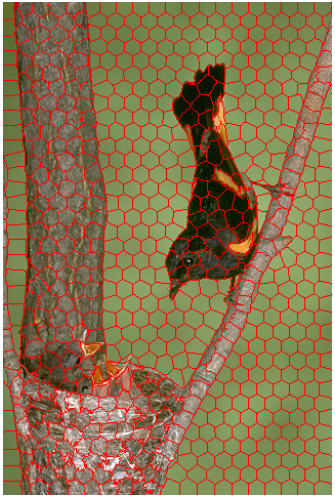
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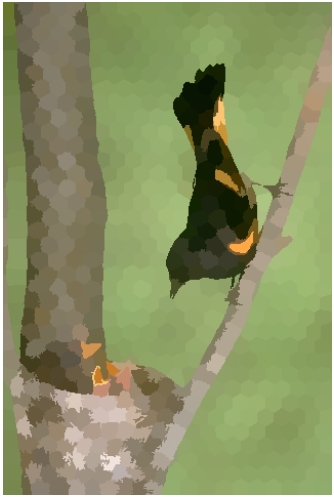
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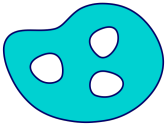
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Simple Connectivity ("Connected Without Holes")

Usually a desired property for superpixels

- Each of the small white regions is simply connected
- Their union is disconnected
- Blue region: connected but not simply connected

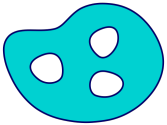


Superpixel approaches to simple connectivity	Examples
• Ignore it & post processing heuristics *(results not directly related to the model)*	[SLIC]
• Respect it & very restricted (e.g. convex polygons)	[Duan et al.]
• Respect only (non-simple) connectivity *(not always an issue)*	[Veksler et al.], [SEEDS]
• Respect it & more free-form, typically:	[Turbopixels], [Chang et al.]

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Respect only (non-simple) connectivity (not always an issue)	[Veksler et al.], [SEEDS]
Respect it & more free-form; typically: • slow • need a large K for good results • favor isotropic superpixels	[Turbopixels], [Chang et al.]

Flexibility in the Overall Shape of a Superpixel

- Many methods: a strong bias to almost-isotropic[2] superpixels
- Poorly suited for elongated structures



SLIC



SLIC

Proposed method Proposed method

- But don't want "too much" flexibility
- Adapt K locally? Slow, K may be too large

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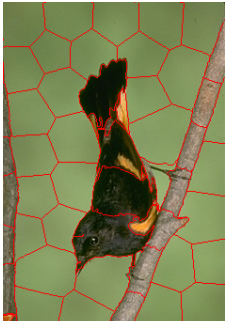
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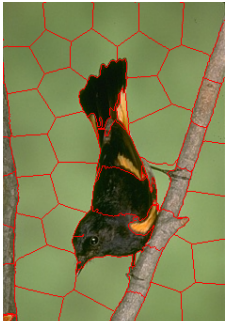
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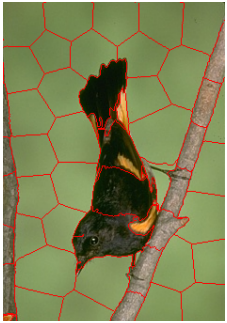
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Problem Statement (First Part)

1. Want a model that
   1. respects simple connectivity
   2. allows for some (inexpensive) flexibility in the shapes of the superpixels

A Model That Respects Simple Connectivity [Chang et al.]

Connectivity-Constrained Gaussian Mixture Model (CC-GMM)

N pixels, K superpixels

Observed pixel i: $\boldsymbol{x}_i = (\boldsymbol{l}_i, \boldsymbol{c}_i)$ = (location, color) $\quad i \in \{1, \ldots, N\}$

Latent label: $z_i = j \iff$ pixel $i \in$ superpixel $j \quad j \in \{1, \ldots, K\}$

GMM: $\boldsymbol{x}_i \sim \sum_{j=1}^{K} w_j \mathcal{N}(\boldsymbol{x}_j; \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)$ $\quad \boldsymbol{\Sigma}_j$ is block diagonal $\quad w_j = \Pr(z_i = j)$

$$\Pr(\overbrace{z_1, \ldots, z_N}^{\text{segmentation}}) = \begin{cases} 0 & \text{if al least one of the superpixels is not simply connected} \\ \text{const} > 0 & \text{otherwise} \end{cases}$$

- Similar to [SLIC[3], Blobworld] except the z_i's are *dependent*
- Good results but slow inference
  Sequential label updates (simple connectivity)
  Expensive splits and merges (shape flexibility)

[3] If all covariances are identical to each other and isotropic

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- Model: GMM with a simple-connectivity constraint
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Modifications:

1. Improved modeling of the spatial covariances
   $\Rightarrow$ better shape flexibility
2. Label updates: parallelize over many pixels (despite the constraint)
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A Bayesian Take on Spatial Covariances

- Unlike [SLIC] or [Chang et al.] we treat the spatial covariances as *latent* and let them *vary across the superpixels* (similar to [Blobworld])
- Unlike [Blobworld], the proposed approach is Bayesian, using an Inverse-Wishart prior
  - Prior: centered on an isotropic covariance
  - Posterior: centered on an anisotropic covariance
- Better flexibility
- Increased robustness to using the "wrong" amount of boundary regularity
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CC-GMM Inference [Chang et al.]

- Q: How to do inference that respects the topological constraint?
- A: Restrict single-label updates to *simple points*[4]

[4] A concept from Digital Topology [Bertrand]

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Simple Point

- Consider the binary case: $K = 2$
- Suppose all superpixels are simply connected
- Let say we want to change the label of some pixel
- Will it preserve simple connectivity? (if so, the pixel is called a simple point)
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But What About Doing it in Parallel?

Problem:

- Simultaneous label updates of simple points can break simple connectivity
- Led to sequential single updates in [Chang et al.]

The proposed solution exploits a simple observation:

- Parallelization over a set of simple points is OK provided no two of which are in the same 3×3 block
- We parallelize updates over $N/9$ pixels (subject to simple points tests)

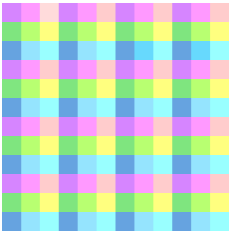
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SLIC



Turbopixels



Veksler e al.



CC-GMM



CC-GMM-SM



Proposed method



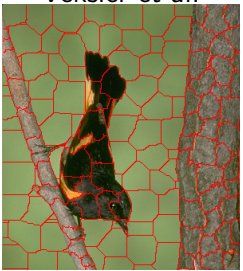
SLIC



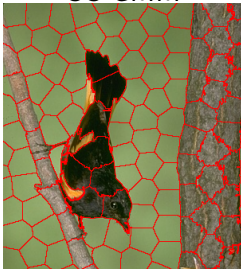
Turbopixels



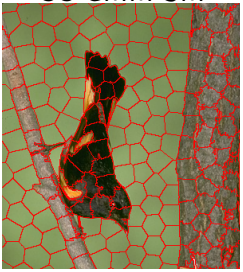
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CC-GMM



CC-GMM-SM

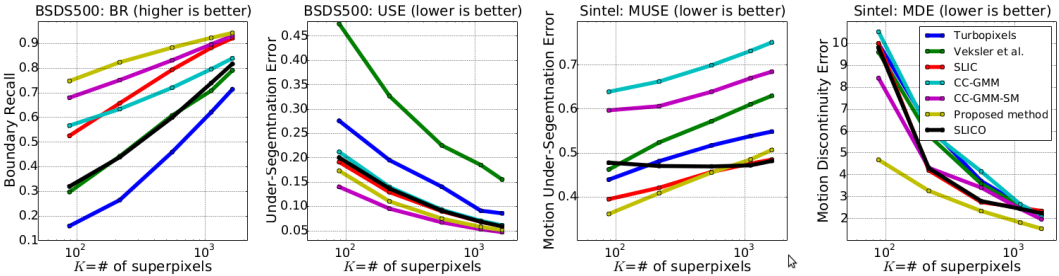


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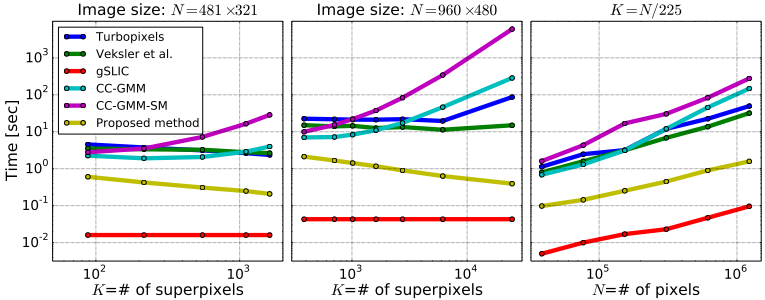
Quantitative Results on Two Public Benchmarks

1) BRSDS500 benchmark; 2) Chemnitz optical-flow-based Sintel benchmark



- Left to right: the proposed method is, #1, #2, #1 (for most K's) and #1
- In the second case, #1 is the (much) slower CC-GMM + Splits/Merges [Chang et al.]

Timings Comparisons



Recent findings (not shown here):

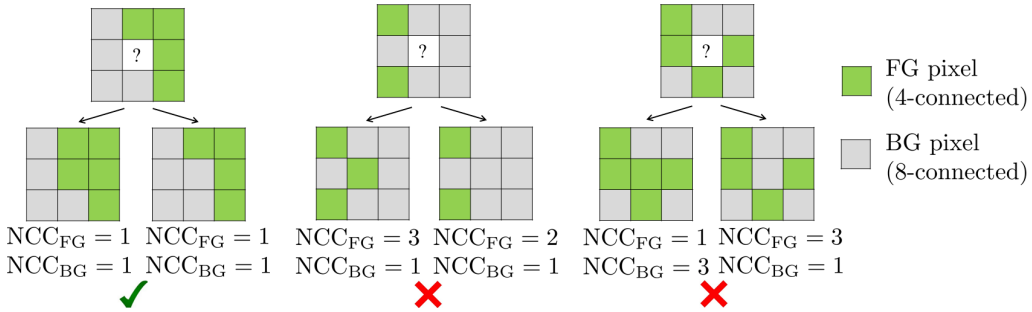
- SEEDS is between the proposed method and SLIC; in quantitative results, however, the proposed method is uniformly better than SEEDS
- GPU Turbopixels has similar timings to the proposed method

Conclusion

- Improved model and faster inference for a GMM over simply-connected superpixels
- For more visual comparisons and our GPU code[5]:
  `http://people.csail.mit.edu/freifeld/publications`
- Some possible extensions:
  - Drop-and-replacement into the video-based pipeline in [Chang et al.]
  - Other features than color
  - Hierarchical models
  - Non-Gaussian distributions for the color/features

[5] Large parts of the code were written by Yixin Li

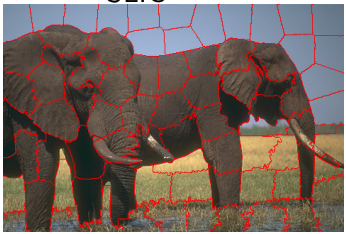
Simple Point Test



Parallelizing the M Step Over Superpixels

- Why does the computing time decrease with an increase in K?
- Higher $K \Rightarrow$ smaller superpixels $\Rightarrow$
  - less terms to sum in the sufficient statistics
  - boundary needs to move a smaller distance $\Rightarrow$ less iterations to converge

SLIC



Turbopixels



Veksler et al.



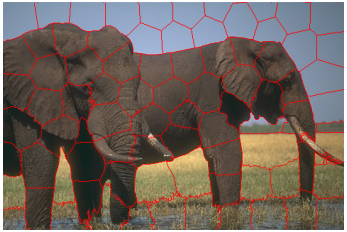
CC-GMM



CC-GMM-SM



Proposed method



SLIC



Turbopixels



Veksler et al.



CC-GMM



CC-GMM-SM



Proposed method



SLIC



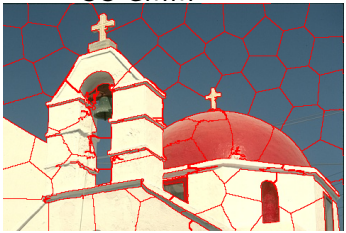
Turbopixels



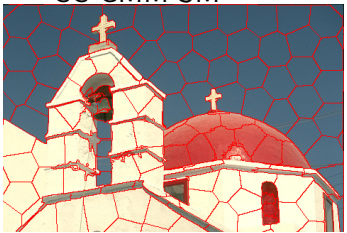
Veksler et al.



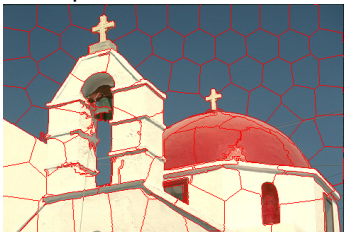
CC-GMM



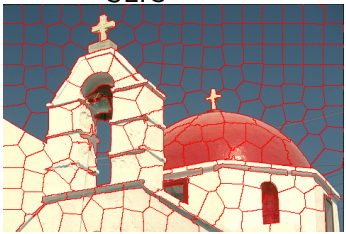
CC-GMM-SM



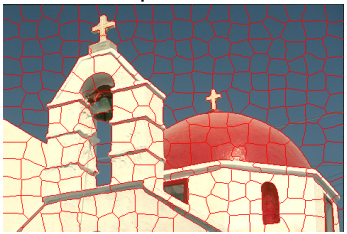
Proposed method



SLIC



Turbopixels



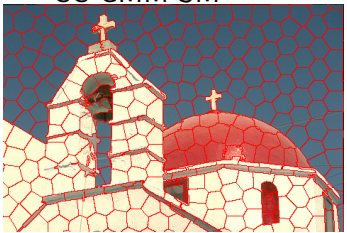
Veksler et al.



CC-GMM



CC-GMM-SM



Proposed method

